CHAPTER 15

Set Theory

It is one of the easiest and most interesting chapters. Almost every year 2-3 questions appear in CAT from this chapter, either in QA or in DI/DS section. The important feature of this chapter is Venn diagrams which play a vital role in the logical reasoning. That's why this chapter is interesting, logical and important in CAT, XLRI, FMS, CET etc. Since we have to solve most of the problems through Venn diagrams, therefore the whole emphasis is given on the theory and concepts of Venn diagrams. Formulae and algebraic theory is not important particularly in CAT, still we have to know some fundamentals of sets and their applications.

15.1 Set

A well defined collection of objects is called a set. The objects in a set are called its members or elements. If a is an element of a set A, then we write $a \in A$ (and say a belongs to A) i.e., 'a' is a member of 'A'. If a does not belong to A, then we write $a \notin A$ By convention the sets are denoted by capital alphabets e.g., $A, B, C, \ldots X, Y, Z$ etc and elements of a set are denoted by $a, b, c, \ldots, x, y, z$ etc.

Important Notations of Standard Sets

N :	the set of natural numbers	Z :	the set of integers
Z^+ :	the set of positive integers	Q :	the set of all rational numbers
Q^+ :	the set of all positive rational numbers	R :	the set of real numbers
R^+ :	the set of all positive real numbers	C :	the set of all complex numbers.

15.2 Methods of Representation of Sets

1. *Roster Method*

In this method a set is described by listing all the elements, separated by commas, within braces { } , *e.g.*

(i) The set of perfect squares upto 100 may be described as {0, 1, 4, 9, 16, 25, 36, 49, 64, 81,100}

(ii) The set of all the days of a week may be described as {Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}.

NOTE The order of elements (*i.e.,* position of elements) is not important. Hence any element can be written anywhere within braces { }.

2. *Set Builder Method*

In this method, a set is described by listing its property in the given manner.

The set $P(x)$ is described as $P = \{ x : P(x) \text{ holds} \}$ or $\{ x \mid P(x) \text{ holds} \}$

Chapter Checklist

- *Set*
- *Methods of Representation of Sets*
- *Types of Sets*
- *Venn Diagrams*
- *Operations on Sets*
- *Algebraic Laws of Sets*
- *Important Results on Number of Elements in Sets*
- *Problems Based on Venn Diagrams*
- *Concept of Maximum or Minimum*
- ***CAT Test***

or $P = \{ x : x \text{ has the property } abc \ldots \}$

which is read as 'the set of all x such that $P(x)$ holds'. The symbol '|' or ':' is read as 'such that', *e.g.,*

1. The set of perfect squares upto 100 can be written as
   $P = \{x : x \text{ is a perfect square and } 0 \le x \le 100, x \in N\}$
   or $P = \{x : x \in N,\ x = n^2,\ 0 \le x < 100, n \in Z\}$
   or $P = \{x^2 : x \in Z\}$
2. The set of all the odd natural numbers can be expressed as $A = \{x : x \in N,\ x = 2n - 1,\ n \in N\}$

15.3 Types of Sets

Singleton Set : A set consisting of a single element is called a singleton set.

e.g. 1. {cat}, {7} etc. 2. $\{x : x + 5 = 7\} = \{2\}$

3. {0} is a singleton set.

Empty Set : A set containing no element at all is called an empty set or a null set or a void set and is denoted by ϕ. In set builder form it is expressed as $\phi = \{\}$

e.g. 1. $\{x : x \in N,\ 9 < x < 10\} = \phi$ 2. $\{x : x \in R,\ x^2 = -8\} = \phi$

NOTE A set containing atleast one element is called a non-empty or non-void set.

Finite Set : If a set has finite number of elements that can be counted or listed *i.e.,* the counting of elements surely comes to an end, is called a finite set, *e.g.,*

1. Set of all the management institutes in the world.
2. Set of all the recognised countries in the world.

Cardinal Number : The number of distinct elements in a finite set is called the 'cardinal number' of the set *e.g.,* $n(A) \to n$ is the number of distinct elements in the set A.

Infinite Set : If the number of elements of a set is infinite *i.e.,* which cann't be counted by natural numbers, is called infinite set in other words if a set is not finite then it must be an infinite set, *e.g.*

1. Set of all points on the arc of a circle
2. Set of all concentric circles with a given centre
3. Number of all lines in a particular plane.
4. $\{x : x \in R,\ 1 < x < 2\}$ and $\{x : x \in I,\ x < 10\}$

Equivalent Sets : Two finite sets A and B are equivalent if their cardinal numbers are same *i.e.* $n(A) = n(B)$

e.g. $A = \{a, e, i, o, u\}$ and $B = \{1, 3, 5, 7, 9\} \Rightarrow n(A) = n(B)$

Equal Sets : Two sets A and B are said to be equal, if every element of A is in B and every element of B is in A and it is denoted as $A = B$. When the elements of A and B are distinct, then $A \neq B$ *i.e.,* A and B are not equal *e.g.,*

1. $A = \{a, e, i, o, u\}$ and $B = \{e, o, i, a, u\} \Rightarrow A = B$
2. $A = \{p, q, r\}, B = \{q, p, r\}$ and $C = \{r, q, p\}$
   $\Rightarrow A = B = C$
3. $A = \{C, A, T\}, B = \{C, A, T, A\}$
   $C = \{A, C, A, A, T\} \Rightarrow A = B = C$

Remark

1. Order of elements in the list of elements is not important *i.e.,* meaningless.
2. The repetition of elements in a set is meaningless.
3. $\phi \neq \{0\} \neq 0$, since ϕ is a set containing no element at all, {0} is a singleton set and 0 is not a set.
   Remember that equivalent sets are not always equal but equal sets are always equivalent. Thus {1, 2, 3,4} and {*W, X, Y, Z*} are equivalent but not equal.

Subsets : Let A and B be two sets such that every element of A is the element of B (*i.e.,* every element of A is in B), then A is called a subset of B and is denoted as-$A \subseteq B$ (Read it as-A is a subset of B)

Superset : If $A \subseteq B$, we say that B is a superset of A and we write $B \supseteq A$. Read it as-B is the superset of A.

e.g. $B = \{2, 4, 6, 8, 10\}$, $A = \{4, 8\} \Rightarrow B \supseteq A$

Proper Subset : If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and denoted as $A \subset B$ *e.g.,*

1. $\{5, 15, 25\} \subset \{5, 10, 15, 20, 25\}$
2. $N \subset W \subset I \subset Q \subset R$

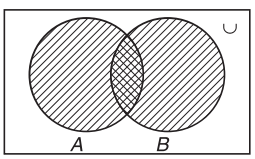
Comparable Sets : Two sets A and B are said to be comparable if either $A \subseteq B$ or $B \subseteq A$.

Results on Subsets

1. The empty set ϕ is a subset of every set.
2. Every set is a subset of itself.
3. The number of all possible subsets of a set containing n elements is 2^n.
4. The number of all proper subsets of a set containing n elements is $(2^n - 1)$.
5. The set of all subsets of a given set A is called the power set of A, denoted by $P(A)$. Thus, if A has n elements $P(A)$ has 2^n elements.

Universal Set : A set that contains all sets in a given context is called the universal set *e.g.,*

1. In a two dimensional coordinate geometry, the set of all points in x-y plane is the universal set.
2. Sets containing natural numbers have a universal set N.
3. If $A = \{1, 2, 4\}$, $B = \{2, 2, 4, 6\}$,
   $C = \{1, 2, 5, 6, 7\}$, then $U = \{1, 2, 3, 4, 5, 6, 7\}$



Power Set : The collection or family of all the subsets of a set A is called the power set of A and is denoted by $P(A)$.

NOTE The power set is always non-empty, because the empty set and the given set itself are the members or elements of the power set.

$A = \{1, 2, 3\}$, then the number of subsets of the $P(A)$ will be $2^3 = 8$

$P(A) = \{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

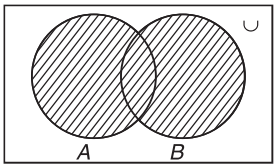
15.4 Venn Diagrams

First of all Swiss mathematician Euler elucidated the idea of representing the sets by points in a closed curve. Later on British mathematician Venn brought this into practice.That's why this whole concept is known as Euler-Venn diagram or simply Venn diagrams. In Venn diagrams the universal set U is represented by points within a rectangle and its subsets are represented by points in closed curves (usually circles) within a rectangle.

15.5 Operations On Sets

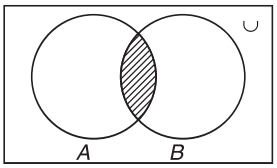
1. **Union of Sets :** If A and B be two sets, then the union of A and B is the set of the all those elements which belong either to A or to B or to both A and B. It is denoted as $A \cup B$ (read as 'A union B') e.g.

   $A = \{2, 3, 5\}$, $B = \{1, 2, 4, 5\}$, $A \cup B = \{1, 2, 3, 4, 5\}$



[The shaded region is representing the union of sets]

2. **Intersection of Sets :** If A and B be two sets then the intersection of A and B is the set of all those elements that belong to both A and B. It is denoted by $A \cap B$ (read as ''A intersection B'')



[The shaded region is representing the intersection of sets]

e.g. $A = \{3, 6, 9, 12, 15, 18\}, B = \{4, 6, 8, 10, 12, 14, 16, 18\}$

$\therefore \quad A \cap B = \{6, 12, 18\}$

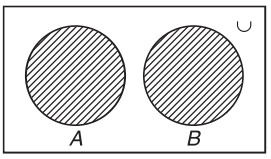
Disjoint Sets : Two sets A and B are said to be disjoint, if $A \cap B = \phi$. If $A \cap B \neq \phi$ then A and B are said to be intersecting or overlapping sets.

e.g. $A = \{1, 3, 5, 7, 9\}$ $B = \{2, 4, 6, 8, 10\}, C = \{4, 8, 12\}$

Here $A \cap B = \phi$ and $B \cap C \neq \phi$

$\therefore$ A and B are disjoint sets

$\therefore$ B and C are intersecting sets.



Sets A and B have no elements common.

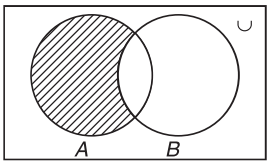
Difference of Sets : The difference of set A and set B (written as $A - B$) is the set of all those elements of A which do not belong to B. Thus, $A - B = \{x : x \in A \text{ and } x \notin B\}$

or $A - B = \{x \in A : x \notin B\}$

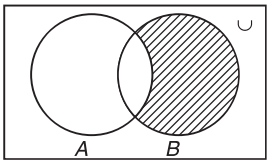
Clearly $x \in A - B \Leftrightarrow x \in A$ and $x \notin B$

e.g. If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10\}$

$A - B = \{1, 3, 5\}$ and $B - A = \{8, 10\}$

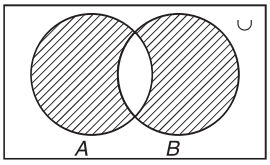


(A – B) is represented by the shaded region



(B – A) is represented by the shaded region

Symmetric difference of two Sets : The symmetric difference of sets A and B is the set $(A - B) \cup (B - A)$ and is denoted by $A \Delta B$

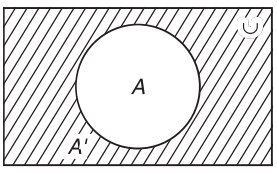


Thus $A \Delta B = (A - B) \cup (B - A) = \{x : x \notin A \cap B\}$

Complement of a Set : If U be the universal set and a set A is such that $A \subset U$ then, the complement of A with respect to U is denoted by A' or A^c or $U - A$ and is defined by the set of all those elements of U which are not in A.

Thus $A' = \{x \in U : x \notin A\}$

Clearly $x \in A' \Leftrightarrow x \notin A$



[A' is shown by the shaded region]

e.g. $U = \{2, 3, 5, 7, 11, 13 \ldots\}$, $A = \{2\}$

$A' = U - A = \{3, 5, 7, 11, 13, \ldots\}$

15.6 Algebraic Laws of Sets

1. **Idempotent Laws :** For any set A, we have
   (i) $A \cup A = A$ (ii) $A \cap A = A$
2. **Identity Laws :** For any set A, we have
   (i) $A \cup \phi = A$ (ii) $A \cap U = A$
3. **Commutative Laws :** For any two sets A and B, we have
   (i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$
4. **Associative Laws :** If A, B and C are any three sets, then
   (i) $(A \cup B) \cup C = A \cup (B \cup C)$
   (ii) $A \cap (B \cap C) = (A \cap B) \cap C$
5. **Distributive Laws :** If A, B and C are any three sets, then
   (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
6. **De-Morgan's Laws :** If A and B are any two sets, then
   (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$

15.7 Important Results on Operations on Numbers

1. $A - B = A \cap B'$
2. $B - A = B \cap A'$
3. $A - B = A \Leftrightarrow A \cap B = \phi$
4. $(A - B) \cup B = A \cup B$
5. $(A - B) \cap B = \phi$
6. $A \subseteq B \Leftrightarrow B' \subseteq A'$
7. $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

15.8 Important Results on Number of Elements in Sets

If A, B and C are finite sets, and U be the finite universal set then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B) = n(A) + n(B) \Leftrightarrow A, B$ are disjoint non-void sets.

(iii) $n(A - B) = n(A) - n(A \cap B)$ *i.e.,* $n(A - B) + n(A \cap B) = n(A)$

(iv) $n(A \Delta B) =$ Number of elements which belong to exactly one of A or $B = n(A) + n(B) - 2n(A \cap B)$

(v) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

(vi) number of elements in exactly two of the sets $A, B, C = n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$

(vii) number of elements in exactly one of the sets $A, B, C = n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(B \cap C) - 2n(A \cap C) + 3n(A \cap B \cap C)$

(viii) $n(A' \cup B') = n[(A \cap B)'] = n(U) - n(A \cap B)$

(ix) $n(A' \cap B') = n[(A \cup B)'] = n(U) - n(A \cup B)$

Introductory Exercise 15.1

1. If $A = \{S\}$, which of the following is a correct statement ?
(a) $S \in A$ (b) $S \subset A$ (c) $\{S\} \in A$ (d) $A = S$

2. Consider the following statements :
1. Any set A is comparable with itself
2. $\{0\}$ is a singleton set
3. $\{\phi\}$ is an empty set.
Of these statements, the correct ones are :
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

3. The total number of elements in the power set of a set A containing n elements is :
(a) n^2 (b) 2^n
(c) $2^n - 1$ (d) none of these

4. Let $A = \{3, \pi, \sqrt{2}, -5, 3 + \sqrt{7}, 2/7\}$. The subset of A containing all the elements from it which are irrational numbers is :
(a) $\{\pi, \sqrt{2}, 3 + \sqrt{7}\}$ (b) $\{3, \pi, 2/7, -5, 3 + \sqrt{7}\}$
(c) $\{3, 2/7, -5\}$ (d) $\{3, -5\}$

5. The set $\{x : (x - 3)(x - 5) > 0\}$ is equal to :
(a) $\{x : 3 < x < 5\}$ (b) $\{x : x < 3\} \cup \{x : x < 5\}$
(c) $\{x : x < 3\} \cup (x : x > 5\}$ (d) none of these

6. If A is the set of all integral multiples of 3 and B is the set of all integral multiples of 5, then $A \cap B$ is the set of all integral multiples of :
(a) 3 + 5 (b) 5 – 3
(c) GCD (3, 5) (d) LCM (3, 5)

7. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n-1) : n \in N$, then precisely :
(a) $X \subset Y$ (b) $X \subseteq Y$
(c) $X = Y$ (d) $X \supseteq Y$

8. The number of elements of the set $\{x : x \in N, x^2 = 1\}$, where N is the set of all natural numbers is :
(a) 2 (b) 1 (c) 0 (d) 3

9. If $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$, then $A \cap B$ is :
(a) $\{1, 2, 3\}$ (b) $\{5, 6, 7\}$
(c) $\{4\}$ (d) ϕ

10. If $A = \{1, 2, 3, ... 9\}$, $B = \{2, 4, 6, 7, 8\}$ and $C = \{3, 4, 5, 8, 9, 10\}$, then $(A - B) \cup C$ is :
(a) $\{1, 3, 4, 5, 8, 9, 10\}$ (b) $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
(c) $\{2, 4, 6, 7, 8\}$ (d) $\{1, 3, 4, 5, 8, 9\}$

11. $A \cup B = A$, then :
(a) $A \subset B$ (b) $B \subset A$
(c) $A \not\subset B$ and $B \not\subset A$ (d) none of these

12. If A has 3 elements and B has 6 elements, then the minimum number of elements in $A \cup B$ is :
(a) 3 (b) 6
(c) 9 (d) 18

13. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set and $A = \{2, 4, 6\}$, $B = \{1, 3, 7\}$, then $A^c \cap B^c$ is equal to :
(a) $\{2, 4, 5, 6, 8, 9, 10\}$ (b) $\{1, 3, 5, 7, 8, 9, 10\}$
(c) X (d) $\{5, 8, 9, 10\}$

14. If A, B, C are any three sets, then :
(a) $A - (B \cup C) = (A - B) \cup (A - C)$
(b) $A - (B \cup C) = (A - B) \cap (A - C)$
(c) $A - (B \cap C) = (A - B) \cup (A - C)$
(d) $A - (B \cup C) = (A \cup B) - (A \cup C)$

15. Consider the following statements :
For any two sets A and B
(1) $(A - B) \cup B = A$ (2) $(A - B) \cup A = A$
(3) $(A - B) \cap B = \phi$ (4) $A \subseteq B \Rightarrow A \cup B = B$
Of these statements :
(a) 1, 2, 3 are correct (b) 2, 3, 4 are correct
(c) 1, 3, 4 are correct (d) 1, 2, 4 are correct

16. If X and Y are two sets and X' denotes the complement of X, then $X \cap (X \cup Y')$ equals :
(a) X (b) Y (c) ϕ (d) none

17. If $B \subseteq A$, which of the following statements is correct?
(a) $A \cup B = B \Rightarrow A = B$ (b) $A \cap B = B \Rightarrow A \supseteq B$
(c) $A - B = A \Rightarrow A = B$ (d) $B - A = B \Rightarrow A = B$

18. If A, B and C be any three sets, then $A \cup (B \cap C)$ is the same as :
(a) $(A \cap C) \cup (B \cap C)$ (b) $(A \cup B) \cap (A \cup C)$
(c) $(A \cup B) \cap (B \cup C)$ (d) $(A \cup C) \cap (B \cup C)$

19. $A - (B \cup C)$ equals :
(a) $(A - B) \cup (A - C)$ (b) $(A - B) \cap (A \cap C)$
(c) $(A - B) \cup C$ (d) $(A - C) \cup B$

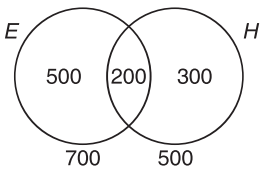
15.9 Problems Based on Venn Diagrams

Venn Diagram Containing Two Circles

Exp. 1) In a group of 1000 people 700 can speak English and 500 can speak Hindi. If all the people speak atleast one of the two languages, find :
(a) how many can speak both the languages?
(b) how many can speak exactly one language?

Solution

(a) $n(E \cup H) = n(E) + n(H) - n(E \cap H)$
$1000 = 700 + 500 - n(E \cap H)$



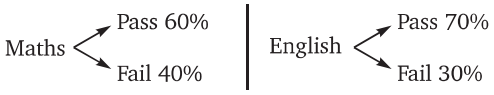
$\Rightarrow \quad n(E \cap H) = 200$

(b) People speaking exactly one language
$n(E \cup H) - n(E \cap H) = 1000 - 200 = 800$

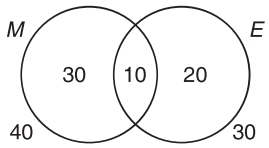
Alternatively People speaking exactly one language
$= 500 + 300 = 800$

Exp. 2) In an exam 60% of the candidates passed in Maths and 70% candidates passed in English and 10% candidates failed in both the subjects. If 300 candidates passed in both the subjects. The total number of candidates appeared in the exam, if they took test in only two subjects viz-Maths and English.

Solution



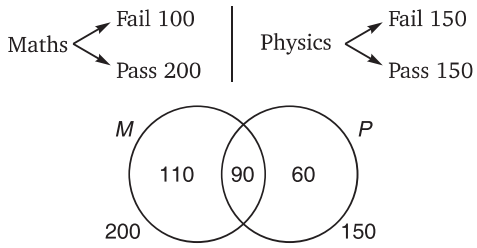
Fail in both subjects = 10%



Failed number of candidates
$\therefore$ Total number of candidates failed in atleast one subject
$= 30 + 10 + 20 = 60\%$
$\therefore$ Total number of candidates passed in both the subjects
$= 100 - 60 = 40\%$
Let total number of candidates appeared in exam $= x$
$\therefore \quad \frac{40x}{100} = 300$
$\Rightarrow \quad x = 750$
Hence, total 750 candidates appeared in exam.

Exp. 3) A class took two tests, one in Physics and other in Maths. $\left(\frac{1}{3}\right)^{rd}$ class failed in Maths and $\frac{1}{2}$ class failed in Physics. 60% of those passed in Physics also passed in Maths. If 300 students took both the tests then how many students failed in both?

Solution



Total number of students passed in atleast one subject

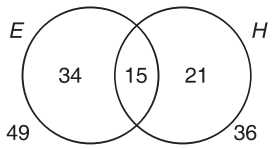
$= 110 + 90 + 60 = 260$

Therefore number of students failed in both the subjects

$= 300 - 260 = 40$

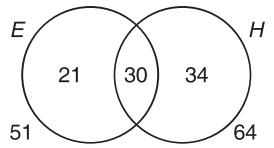
Exp. 4) In an exam 49% candidates failed in English and 36% failed in Hindi and 15% failed in both subjects. If the total number of candidates who passed in English alone is 630. What is the total number of candidates appeared in exam?

Solution



Number of candidates failed in atleast one subject

$= 34 + 15 + 21 = 70\%$



$\therefore$ Number of candidates passed in both the subjects

$= 100 - 70 = 30\%$

Number of candidates passed in English only

$= 51 - 30 = 21\%$

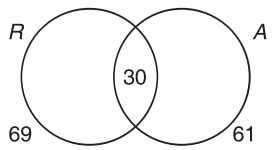
Let the total number of candidates be x, then $\frac{21x}{100} = 630$

$\Rightarrow \quad x = 3000$

Alternatively Since 21% candidates failed in Hindi only it means they are pass in only English. Hence, number of candidates who are passed only in English = 21%

Exp. 5) In a survey it was found that 69% of Indians like romantic movies and 61% like action movies. What percentage of Indians like both type of movies?

Solution $n(R \cap A) = (69 + 61) - 100$

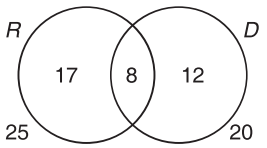


NOTE In general we cannot determine the solution but this can be done if we assume that all the Indians like atleast one of these two types of movies.

Exp. 6) In a certain city only two cellular (mobile) phone services are available. One is Radiance and other is DSML. It is known that 25% of the city population uses the services of Radiance and 20% uses the services of DSML, while 8% uses both Radiance and DSML. It is also known that 30% of those who uses Radiance services but not services of DSML have prepaid cards and 40% of those who uses DSML but not Radiance have prepaid cards while 50% of those who uses the services of both Radiance and DSML have prepaid cards. What percentage of the population have prepaid cards?

(a) 12·5% (b) 10% (c) 13·9% (d) 20%

Solution

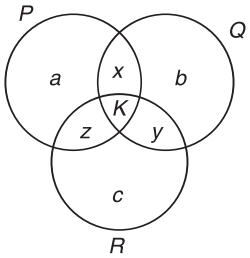


$\therefore$ Percentage of people have prepaid cards

$= (30\ \%\ \text{of}\ 17) + (40\%\ \text{of}\ 12) + (50\%\ \text{of}\ 8) = 13.9\%$

Ven Diagrams Containing Three Circles

Explanation of the concept using an example



Let P, Q, R be three books, then

$a \rightarrow$ number of people reading only P

$b \rightarrow$ number of people reading only Q

$c \rightarrow$ number of people reading only R

$x \rightarrow$ number of people reading P and Q both but not R

$y \rightarrow$ number of people reading Q and R both but not P

$z \rightarrow$ number of people reading P and R both but not Q

$k \rightarrow$ number of people reading all the three books

$x + k \rightarrow$ number of people reading P and Q both

$y + k \rightarrow$ number of people reading Q and R both

$z + k \rightarrow$ number of people reading P and R both

$a + x + k + z \rightarrow$ number of people reading P

$b + x + k + y \rightarrow$ number of people reading Q

$c + y + k + z \rightarrow$ number of people reading R

$a + x + b \rightarrow$ number of people reading P or Q but not R

$b + y + c \rightarrow$ number of people reading Q or R but not P

$c + z + a \rightarrow$ number of people reading P or R but not Q.

$a + b + c \rightarrow$ number of people reading only one book

$x + y + z \rightarrow$ number of people reading only two books

$k \rightarrow$ number of people reading all of three books

$(a + b + c) + (x + y + z) + (k) \rightarrow$ number of people reading atleast one book $(x + y + z) + (k) \rightarrow$ number of people reading atleast two books $(a + b + c) + (x + y + z) + (k) \rightarrow$ total number of readers.

Finding the Values of Components

$P + Q + R = (a + x + k + z) + (b + x + k + y) + (c + y + k + z)$

$P + Q + R = (a + b + c) + 2(x + y + z) + 3k$

$P + Q + R = \alpha + 2\beta + 3\gamma$

where $\alpha = a + b + c, \quad \beta = x + y + z, \quad \gamma = k$

Again $(x + k) + (y + k) + (z + k) = (x + y + z) + 3k = \beta + 3\gamma$

$\therefore \quad (\alpha + 2\beta + 3\gamma) - (\beta + 3\gamma) = \alpha + \beta$

and $\quad (\alpha + \beta + \gamma) - (\alpha + \beta) = \gamma$

NOTE 'And' means intersection, 'or' means union

Exp. 1) In a group of 100 children, 32% play hockey, 64% play football and 40% play cricket. 14% play hockey and football, 15% play football and cricket, 13% play hockey and cricket. Only 6% play all the three games.

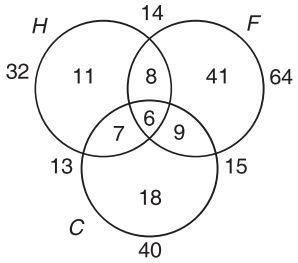
(a) How many children play only one game?

(b) How many children play atleast two game?

(c) How many children play hockey or football but not circket?

Solution (a) $11 + 41 + 18 = 70$ (b) $(8 + 7 + 9) + 6 = 30$

(c) $(11 + 41) + 8 = 60$



Exp. 2) In a hostel of 100 students capacity, 42 are science graduate, 34 are law graduate and 58 are commerce graduate. 8 students are science and law graduate, 13 students are law and commerce graduate, 18 students are commerce and science graduate. How many students are only science graduate?

Solution $\alpha + \beta + \gamma = 100; \alpha + 2\beta + 3\gamma = 42 + 34 + 58 = 134$

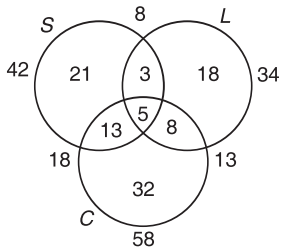
$\beta + 3\gamma = 8 + 13 + 18 = 39$

$\alpha + \beta = (\alpha + 2\beta + 3\gamma) - (\beta + 3\gamma) = 134 - 39 = 95$

$\therefore \quad \gamma = (\alpha + \beta + \gamma) - (\alpha + \beta) \Rightarrow \gamma = 100 - 95 = 5$

$\therefore$ Number of students who are only science graduate

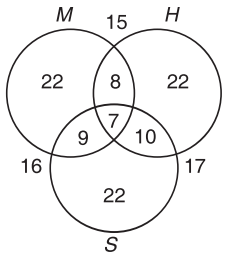
$= 42 - [(18 + 8) - 5] = 42 - (13 + 3 + 5) = 21$



Exp. 3) In a premier B-school having 100 students in the MBA final year, 15 students specializes in Marketing and HR. 17 students specializes in HR and Systems, 16 students specializes in Marketing and Systems. The number of students specializes in only one field is equal and is 22 in each field of specialization. How many students specializes in all three fields viz- Marketing, HR and Systems?

Solution $\alpha = a + b + c = 22 + 22 + 22 = 66$

$\therefore \quad \beta + \gamma = 100 - \alpha = 100 - 66 = 34$



Again $\quad \beta + 3\gamma = x + y + z + 3k = 15 + 16 + 17 = 48$

$\therefore \quad (\beta + 3\gamma) - (\beta + \gamma) = 2\gamma = 48 - 34 = 14 \quad \Rightarrow \gamma = 7$

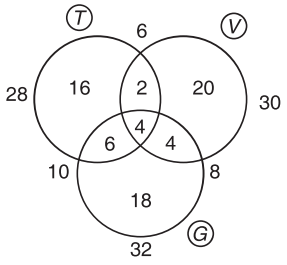
Hence, 7 students specializes in all the three fields.

Exp. 4) In a music school 28 students learn trumpet, 30 students learn violin and 32 students learn guitar. 6 students learn trumpet and violin, 8 students learn violin and guitar, 10 students learn guitar and trumpet. The number of students who learn only one instrument is 54. Also 20 students learn only violin. Every student learn atleast one instrument out of the three instruments.

(a) Find the number of students in the music school.

(b) Find the number of students who learn trumpet and guitar but not violin?

Solution



$x + b + k + y = 30$

$b = 20$

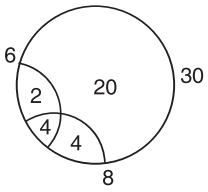
$\Rightarrow \quad x + k + y = 10$

But, $\quad (x + k) + (y + k) = 14$

$\Rightarrow \quad (x + y + k) + k = 14$

$\Rightarrow \quad 10 + k = 14$

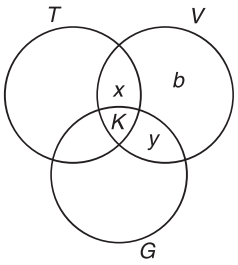
$\Rightarrow \quad k = 4$



(a) Number of students in music school

$= (16 + 20 + 18) + (2 + 4 + 6) + 4 = 70$

(b) Number of students who learn trumpet and guitar but not violin = 6



Exp. 5) Out of 180 students in a class 7 students failed in all the three subjects viz., Physics, Chemistry, Maths. Only 144 students pass in only one subject and 21 students pass in only two subjects.

(a) How many students passed in all the three subjects?

(b) How many students passed in atleast two subjects?

(c) How many students passed in at most two subjects but atleast in one subject?

Solution $\alpha = a + b + c = 144 \Rightarrow \beta = x + y + z = 21$

$\alpha + \beta + \gamma = 180 - 7 = 173 \quad \therefore \gamma = (\alpha + \beta + \gamma) - (\alpha + \beta)$

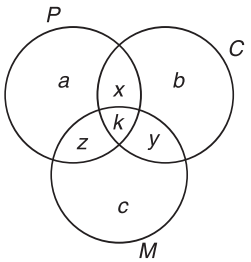
$\Rightarrow \quad \gamma = (173) - (144 + 21) \quad \Rightarrow \gamma = 8 (\gamma = k)$

(a) 8

(b) Atleast two subjects $= \beta + \gamma = 21 + 8 = 29$

(c) Atleast one subject but atmost two subjects.

$= \alpha + \beta = 144 + 21 = 165.$



Exp. 6) In a marriage party total 278 guests were present. 20 guests took Pepsi and Dew, 23 guests took Dew and Sprite and 21 guests took Pepsi and Sprite and 9 guests took all the three cold drinks viz. Pepsi, Sprite and Dew. It is also known that there were equal number of bottles of each of three kinds viz. Pepsi, Dew and Sprite

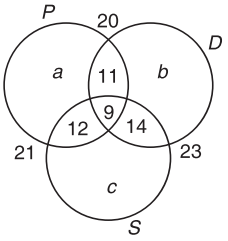
(a) How many guests took only one bottle colddrink?

(b) How many guests enjoyed Dew or Sprite but not Pepsi?

(c) How many guests enjoyed Pepsi and Dew but not Sprite?

Solution Number of guests $= (a + b + c) + (11 + 12 + 14) + (9)$

$= 278$



Number of bottles $= (a + b + c) + 2(11 + 12 + 14) + 3(9)$

$= [(a + b + c) + (11 + 12 + 14) + (9)] + (11 + 12 + 14) + 2(9)$

$= 278 + 55 = 333$

$\therefore \quad$ Each kind of bottles $= \dfrac{333}{3} = 111$

$\because \quad a = 111 - (11 + 9 + 12) = 79$

$b = 111 - (11 + 9 + 14) = 77$

$c = 111 - (12 + 9 + 14) = 76$

(a) Number of guests who took only one bottle $= a + b + c$

$= (79 + 77 + 76) = 232$

(b) Dew or Sprite but not Pepsi $= b + 14 + c$

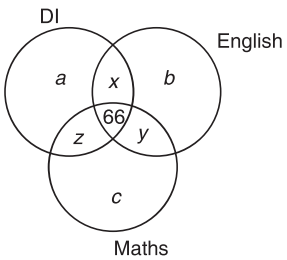
$= 77 + 14 + 76 = 167$

(c) Dew and Pepsi but not Sprite $= 20 - 9 = 11$

Exp. 7) 250 students appeared in MOCK CAT consisting of three sections viz. Maths, Data Interpretation (DI) and English. 20 students did not qualify any of the three sections and 66 students qualified all the three sections, 24 students did not qualify DI only. 12 did not qualify DI and Maths only.

Those who qualified Maths only exceeded those who did not qualify any of the section by 16. Those who did not qualify Maths only exceeded those who did not qualify only DI and Maths by 8. Those who did not qualify English only were twice the number of students who did not qualify Maths only. Find the number of students who has qualified atleast two sections.

Solution Total students = 250



Number of students who did not qualify any of the three sections = 20

∴Number of students who qualified atleast one section
= 250 – 20 = 230

Hint $y = 24$ (It means 24 students must have qualified English and Maths) $b = 12$ (It means they have qualified neither DI nor Maths but English only)

$c = 36 = (20 + 16) \Rightarrow x = 20 = (12 + 8) \Rightarrow z = 40 = (20 \times 2)$

$\therefore \quad (a + b + c) + (x + y + z) + 66 = 230$

$(a + 12 + 36) + (20 + 24 + 40) + 66 = 230 \Rightarrow a = 32$

but the required answer = $(x + y + z) + 66 = 84 + 66 = 150$

15.10 Concept of Maxima and Minima

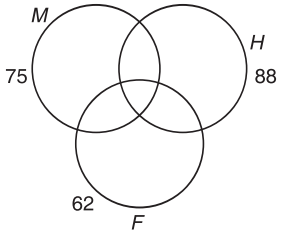
Type 1. In a Business School there are 3 electives and atleast one elective is compulsory to opt. 75% students opted for Marketing, 62% students opted for Finance and 88% students opted for HR. A student can have dual or triple specialization.

(a) What is the minimum number of students that are specialize in all three streams?

(b) What is the maximum number of students that can specialize in all three streams?

Solution

(a) 75% students opted for Marketing, it means atleast 25% students opted for HR or Finance or HR and Finance both.

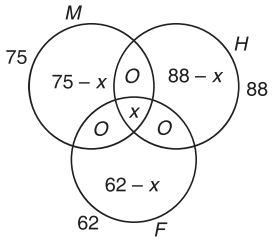


Similarly atleast 12% students opted for Marketing or Finance or Marketing and Finance both. Again atleast 38% students opted for Marketing or HR or Marketing and HR both.

$\therefore \qquad 25 + 12 + 38 = 75\%$

It means that if there is no intersection in these three courses,75% would be maximum number of students in M, H, F alone or $(M \cap H), (M \cap F), (H \cap F)$. Thus it gives atleast 25% = (100 – 75) students in all 3 departments.

(b) $(75 - x) + (88 - x) + (62 - x) + x = 100 \Rightarrow x = 62.5\%$

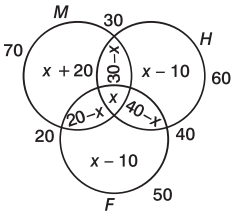


Type 2. In a Business school there are 3 electives offered to the students, where students have a choice of not choosing any electives. 70% students opted for Marketing, 60% students opted for HR and 50% students opted for Finance. 30% students opted for Marketing and HR both, 40% opted for HR and Finance both, 20% opted for Marketing and Finance both.

(a) What is the minimum number of students opted for all 3 electives?

(b) What is the maximum number of students opted for all 3 electives?

Solution Total number of students = $70 + (x - 10) + (40 - x) + (x - 10) = 90 + x$



(a) The minimum value of x so that none of the categories becomes negative = 10

(b) The maximum value of x so that none of the categories becomes negative = 20

Type 3. In a hostel there are 250 students. 120 watch news channels, 80 watch sports channels and 90 watch music channels. Each channel telecasts the content as per their nature. 50 students watch both news and sports channel 60 students watch both sports and music channel and 65 students watch both news and music channels.

(a) What is the minimum number of students who watch atleast one of the given channel?

(b) What is the maximum number of students who watch atleast one of the given channels?

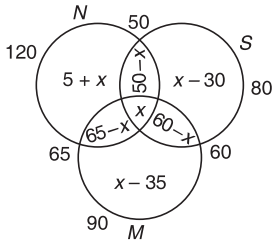
(c) What is the minimum number of students who watch all the three types of channels?

(d) What is the maximum number of students who watch all the three types of channels?

Solution Since the number of students cannot be negative

$$\therefore \quad \left.\begin{aligned} x - 30 &\geq 0 \\ x - 35 &\geq 0 \end{aligned}\right\}$$

$$\text{Also} \quad \left.\begin{aligned} 50 - x &\geq 0 \\ 60 - x &\geq 0 \\ 65 - x &\geq 0 \end{aligned}\right\}$$



Total number of students playing atleast one game
$= 120 + (x - 30) + (60 - x) + (x - 35) = 115 + x$

(a) Minimum number of students watching atleast one channel = 115 + 35 = 150

(b) Maximum number of students watching atleast one channel = 115 + 50 = 165

(c) Minimum number of students watching all the 3 channels = 35

(d) Maximum number of students watching all the 3 channels = 50